LNP™ THERMOTUF™ COMPOUND OF006IXP

DESCRIPTION

LNP THERMOTUF OF006IXP is a compound based on linear Polyphenylene Sulfide (PPS) resin containing glass fiber. Added features of this material include: high impact resistance, high dimensional stability and low warpage.

TYPICAL PROPERTY VALUES

Revision 20210812

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
MECHANICAL			
Tensile Modulus, 1 mm/min	10300	MPa	ISO 527
Tensile Stress, break, 5 mm/min	140	MPa	ISO 527
Tensile Strain, break, 5 mm/min	2.4	%	ISO 527
Flexural Modulus, 2 mm/min	9300	MPa	ISO 178
Flexural Strength, 2 mm/min	205	MPa	ISO 178
Tensile Modulus, 5 mm/min	10500	MPa	ASTM D638
Tensile Stress, brk, Type I, 5 mm/min	143	MPa	ASTM D638
Tensile Strain, brk, Type I, 5 mm/min	2.6	%	ASTM D638
Flexural Modulus, 1.3 mm/min, 50 mm span	9800	MPa	ASTM D790
Flexural Strength, 1.3 mm/min, 50 mm span	210	MPa	ASTM D790
IMPACT			
Izod Impact, notched 80*10*4 +23°C	16.5	kJ/m²	ISO 180/1A
Izod Impact, notched 80*10*4 -40°C	12	kJ/m²	ISO 180/1A
Izod Impact, unnotched 80*10*4 +23°C	50	kJ/m²	ISO 180/1U
Izod Impact, unnotched 80*10*4 -40°C	55	kJ/m²	ISO 180/1U
Charpy 23°C, V-notch Edgew 80*10*3 sp=62mm	17	kJ/m²	ISO 179/1eA
Charpy 23°C, Unnotch Edgew 80*10*4 sp=62mm	55	kJ/m²	ISO 179/1eU
Izod Impact, notched, 23°C	170	J/m	ASTM D256
Izod Impact, unnotched, 23°C	850	J/m	ASTM D4812
THERMAL			
HDT/Bf, 0.45 MPa Flatw 80*10*4 sp=64mm	278	°C	ISO 75/Bf
HDT/Af, 1.8 MPa Flatw 80*10*4 sp=64mm	260	°C	ISO 75/Af
CTE, -40°C to 40°C, flow	1.5E-05	1/°C	ISO 11359-2
CTE, -40°C to 40°C, xflow	4.5E-05	1/°C	ISO 11359-2
CTE, -40°C to 90°C, flow	1.6E-05	1/°C	ISO 11359-2
CTE, -40°C to 90°C, xflow	5.2E-05	1/°C	ISO 11359-2
HDT, 0.45 MPa, 3.2 mm, unannealed	275	°C	ASTM D648
HDT, 1.82 MPa, 3.2mm, unannealed	258	°C	ASTM D648
CTE, -40°C to 90°C, flow	1.5E-05	1/°C	ASTM E831
CTE, -40°C to 90°C, xflow	5.0E-05	1/°C	ASTM E831
PHYSICAL			
Density	1.49	g/cm³	ISO 1183
Water Absorption, (23°C/24hrs)	0.03	%	ISO 62-1
Water Absorption, (23°C/saturated)	0.05	%	ISO 62-1
Moisture Absorption, (23°C/50% RH/24hrs)	0.01	%	ISO 62-4

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
Moisture Absorption, (23°C/50% RH/Equilibrium)	0.01	%	ISO 62-4
Specific Gravity	1.49	-	ASTM D792
Water Absorption, (23°C/24hrs)	0.03	%	ASTM D570
Melt Volume Rate, MVR at 316°C/5.0 kg	10	cm^3/10 min	ASTM D1238
Melt Flow Rate, 316°C/5.0 kgf	15	g/10 min	ASTM D1238
Mold Shrinkage, flow	0.2 – 0.3	%	SABIC method
Mold Shrinkage, xflow	0.4 – 0.5	%	SABIC method
INJECTION MOLDING			
Drying Temperature	120 – 140	°C	
Drying Time	3 – 5	Hrs	
Hopper Temperature	50 – 70	°C	
Nozzle Temperature	310 – 330	°C	
Melt Temperature	310 – 330	°C	
Front - Zone 3 Temperature	310 – 330	°C	
Middle - Zone 2 Temperature	310 – 330	°C	
Rear - Zone 1 Temperature	290 – 320	°C	
Mold Temperature	130 – 160	°C	
Back Pressure	0.2 – 0.5	MPa	
Screw Speed	50 – 100	rpm	

DISCLAIMER